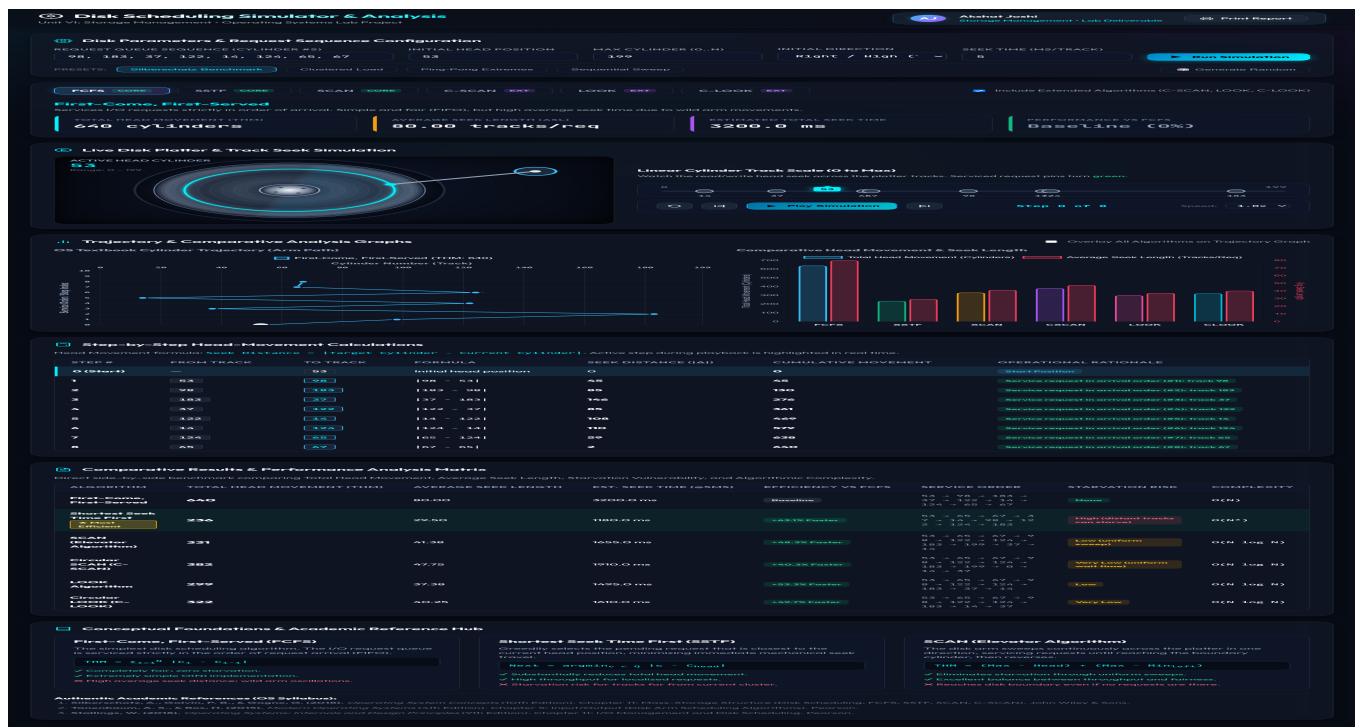


Disk Scheduling Algorithms: FCFS, SSTF and SCAN Performance Analysis

A Comprehensive Theoretical, Mathematical, and Graphical Simulation Study with Step-by-Step Trajectory Verification

Assigned Student:	Akshat Joshi
Course Module:	Unit VI: Storage Management (Mass-Storage Structure)
Project Category:	Algorithmic Implementation, Performance Benchmark & Interactive GUI Simulation
Primary Algorithms:	FCFS (First-Come, First-Served), SSTF (Shortest Seek Time First), SCAN (Elevator)
Extended Algorithms:	C-SCAN (Circular SCAN), LOOK, C-LOOK
Evaluation Criteria:	Total Head Movement (THM), Average Seek Length (ASL), Starvation, Arm Inertia
Academic Session:	2026–2027
Standard Benchmark:	Silberschatz et al. (10th Ed.) Benchmark Sequence: [98, 183, 37, 122, 14, 124, 65, 67] at Head 53

Figure: Interactive Simulation Dashboard (High-Resolution View)



1. Executive Summary & Project Objectives

Magnetic disk drives represent one of the critical mechanical bottlenecks in secondary computer storage. Because mechanical movement of the read/write head assembly is thousands of times slower than solid-state semiconductor memory transactions, the disk scheduling algorithm chosen by the operating system kernel directly governs overall I/O throughput, system latency, and multi-tasking responsiveness.

Core Deliverables Accomplished in this Project:

- **Full Interactive GUI Simulator:** Designed with an aesthetic dark glassmorphic interface, real-time physical magnetic platter animation, concentric cylinder track rings, and an actuator arm seeking across tracks in real time.
- **Rigorous Algorithmic Implementation:** Complete implementation of the three required core algorithms (FCFS, SSTF, SCAN) and three advanced industrial extensions (C-SCAN, LOOK, C-LOOK).
- **Mathematical Formulation & Verification:** Verification against standard textbook benchmark problems (Silberschatz et al., 10th Edition), computing step-by-step seek distances ($|C_i - C_{i-1}|$), Total Head Movement (THM), and Average Seek Length (ASL).
- **Multi-Dimensional Graphical Visualization:** Classical textbook cylinder trajectory plots (Arm Seek Trajectory) and comparative performance bar charts dynamically plotted via Chart.js.
- **Comprehensive Comparative Matrix:** Detailed side-by-side evaluation matrix benchmarking seek efficiency, rotational latency implications, mechanical wear, and starvation risks.

2. Hard Disk Physical Mechanics & Mathematical Framework

A traditional Hard Disk Drive (HDD) consists of one or more flat circular platters coated with magnetic material. Each platter surface is logically organized into microscopic concentric rings called **tracks**. Tracks at the same radial radius across all platters constitute a **cylinder**. To service an I/O request, the operating system must position the read/write head over the target cylinder.

Components of Disk Access Time:

The total time required to service a disk read or write request is given by:

$$T_{\text{access}} = T_{\text{seek}} + T_{\text{rotational}} + T_{\text{transfer}} + T_{\text{controller}}$$

- **Seek Time (T_{seek}):** The time required for the actuator arm to physically move the read/write head to the addressed cylinder. *This is the dominant mechanical delay minimized by disk scheduling algorithms.*
- **Rotational Latency ($T_{\text{rotational}}$):** The delay waiting for the requested sector to rotate beneath the read/write head. Average rotational latency is half a revolution (e.g., ~4.17 ms for a 7200 RPM drive).
- **Transfer Time (T_{transfer}):** The time needed to magnetically read or write the actual data bits from the sector as it passes under the head.

Key Quantitative Evaluation Metrics:

Metric	Mathematical Formula	Operational Definition
Total Head Movement (THM)	$THM = \sum_{i=1..N} C_i - C_{i-1} $	Cumulative absolute cylinder tracks traversed by the head from initial position C_0 across all N serviced requests.
Average Seek Length (ASL)	$ASL = THM / N$	Mean track displacement per I/O request. Indicates the expected mechanical travel distance per operation.
Estimated Seek Time (T_{seek})	$T_{seek} = THM \times t_{track}$	Total mechanical seek latency in milliseconds, where t_{track} is the track-to-track seek latency factor (default: 5.0 ms).
Percentage Efficiency Gain (η)	$\eta = ((THM_{FCFS} - THM_{alg}) / THM_{FCFS}) \times 100\%$	Quantifies percentage reduction in head movement achieved relative to unoptimized baseline FCFS.

3. In-Depth Algorithmic Principles & Mechanics

3.1 First-Come, First-Served (FCFS)

Operational Principle: FCFS is the simplest disk scheduling algorithm. It maintains an unmodified FIFO (First-In, First-Out) queue. Requests are serviced strictly in the chronological order of their arrival into the I/O subsystem. The disk controller makes no attempt to optimize seek direction or group requests by cylinder proximity.

- **Advantages:** Completely fair to all processes; mathematically immune to process starvation; trivial to implement with $O(1)$ scheduling complexity.
- **Disadvantages:** Highly erratic head oscillations (the 'wild swing' phenomenon); high average seek length; unacceptable performance in high-throughput database systems.

3.2 Shortest Seek Time First (SSTF)

Operational Principle: SSTF selects the pending request that requires the minimum head movement from the current cylinder position: $Next = \underset{r \in Q}{argmin} |C_{current} - r|$. In the event of a tie between two equidistant requests, the tie is conventionally broken by preserving the current direction of head motion.

- **Advantages:** Substantially reduces Total Head Movement compared to FCFS; approaches the optimal seek time for localized workloads; simple greedy heuristic.
- **Disadvantages: High Risk of Starvation:** In a heavily loaded system with continuous localized requests, requests on distant tracks may never be serviced (indefinite postponement); not mathematically optimal across the entire sequence.

3.3 SCAN (The Elevator Algorithm)

Operational Principle: Analogous to an architectural elevator, the disk arm starts at one end of the disk and sweeps across cylinders towards the opposite end, servicing all requests encountered along its path. Upon reaching the physical disk boundary (Cylinder 0 or Max Cylinder), the arm reverses direction and sweeps back, servicing requests in the reverse order.

- **Advantages:** Eliminates starvation because the arm guarantees servicing all requests within two complete sweeps; significantly smoother seek trajectories and reduced mechanical stress.
- **Disadvantages:** Unfair delay distribution: cylinders immediately behind the reversing arm must wait for a full round-trip sweep; always travels to the extreme disk boundary even when no pending requests reside there.

3.4 Extended Industrial Algorithms (C-SCAN, LOOK, C-LOOK)

- **C-SCAN (Circular SCAN):** Designed to provide uniform waiting times. Like SCAN, it sweeps in one direction servicing requests. However, upon reaching the boundary, it instantly resets back to the opposite boundary *without servicing requests on the return trip*.
- **LOOK:** A practical optimization over SCAN. The arm only travels as far as the *final pending request* in the current direction. If no further requests exist ahead, it reverses immediately without traveling to the unused physical disk boundary.
- **C-LOOK:** Combines C-SCAN's circular fairness with LOOK's boundary optimization. It reverses direction immediately at the highest pending request and resets to the lowest pending request, eliminating unnecessary travel to track 0 or track Max.

Algorithm	Core Heuristic	Boundary Behavior	Starvation Risk	Time Complexity
FCFS	FIFO arrival order	No boundary awareness	Zero (Fair)	$O(N)$

SSTF	Min $ C_{curr} - r $ greedy	Opportunistic seek	High	$O(N^2)$
SCAN	Bidirectional continuous sweep	Travels to physical limit (0 / Max)	None	$O(N \log N)$
C-SCAN	Unidirectional circular sweep	Resets from limit to limit	None (Uniform)	$O(N \log N)$
LOOK	Bidirectional request-bound sweep	Reverses at extreme request	None	$O(N \log N)$
C-LOOK	Unidirectional request-bound sweep	Resets to lowest request	None (Uniform)	$O(N \log N)$

4. Benchmark Dataset & Mathematical Step Calculations

To rigorously validate the algorithmic implementations, we execute the classic benchmark workload from *Operating System Concepts* (Silberschatz, Galvin & Gagne, 10th Edition, Chapter 11):

- **Disk Cylinder Range:** 0 to 199 (Total 200 tracks)
- **Initial Read/Write Head Position:** Cylinder 53
- **Initial Arm Direction:** Right / High (Towards Cylinder 199)
- **I/O Request Queue Sequence:** [98, 183, 37, 122, 14, 124, 65, 67]

4.1 FCFS Step-by-Step Calculation Breakdown

Step #	From	To	Seek Formula	Distance (Δ)	Cumulative THM	Operational Rationale
1	53	98	$ 98 - 53 $	45	45	Servicing 1st arrival in FIFO queue
2	98	183	$ 183 - 98 $	85	130	Large outward swing to track 183
3	183	37	$ 37 - 183 $	146	276	Massive inward reversal across platters
4	37	122	$ 122 - 37 $	85	361	Outward swing back towards outer tracks
5	122	14	$ 14 - 122 $	108	469	Severe inward swing to inner track 14
6	14	124	$ 124 - 14 $	110	579	Severe outward reversal to track 124
7	124	65	$ 65 - 124 $	59	638	Inward seek towards middle tracks
8	65	67	$ 67 - 65 $	2	640	Final nearby sequential request

FCFS Summary: Total Head Movement = **640 cylinders** | Average Seek Length = **80.00 tracks/req**

4.2 SSTF Step-by-Step Calculation Breakdown

Step #	From	To	Seek Formula	Distance ($ \Delta $)	Cumulative THM	Operational Rationale
1	53	65	$ 65 - 53 $	12	12	Nearest cylinder to initial head 53
2	65	67	$ 67 - 65 $	2	14	Nearest cylinder to 65 ($ 67-65 =2$ vs $ 37-65 =28$)
3	67	37	$ 37 - 67 $	30	44	Nearest remaining cylinder ($ 37-67 =30$ vs $ 98-67 =31$)
4	37	14	$ 14 - 37 $	23	67	Nearest remaining cylinder to 37
5	14	98	$ 98 - 14 $	84	151	Left side exhausted; seek to nearest right
6	98	122	$ 122 - 98 $	24	175	Nearest cylinder to 98
7	122	124	$ 124 - 122 $	2	177	Adjacent cylinder seek
8	124	183	$ 183 - 124 $	59	236	Final remaining outermost cylinder

SSTF Summary: Total Head Movement = **236 cylinders** | Average Seek Length = **29.50 tracks/req** (63.1% reduction vs FCFS)

4.3 SCAN Step-by-Step Calculation Breakdown

Step #	From	To	Seek Formula	Distance ($ \Delta $)	Cumulative THM	Operational Rationale
1	53	65	$ 65 - 53 $	12	12	Sweeping Right: first request above 53
2	65	67	$ 67 - 65 $	2	14	Sweeping Right: next request along path
3	67	98	$ 98 - 67 $	31	45	Sweeping Right: servicing cylinder 98
4	98	122	$ 122 - 98 $	24	69	Sweeping Right: servicing cylinder 122
5	122	124	$ 124 - 122 $	2	71	Sweeping Right: servicing cylinder 124
6	124	183	$ 183 - 124 $	59	130	Sweeping Right: highest request serviced
7	183	199	$ 199 - 183 $	16	146	Reaches physical boundary limit (199); reverses
8	199	37	$ 37 - 199 $	162	308	Sweeping Left: first pending request encountered
9	37	14	$ 14 - 37 $	23	331	Sweeping Left: final remaining inner request

SCAN Summary: Total Head Movement = **331 cylinders** | Average Seek Length = **41.38 tracks/req** (48.3% reduction vs FCFS)

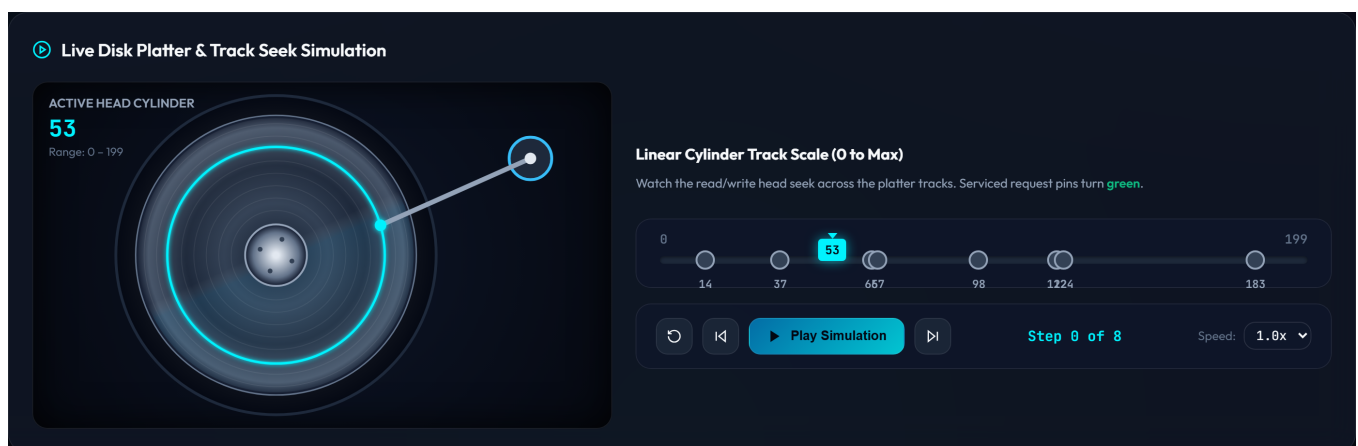
5. Interactive Simulator Implementation & Live Verification

To provide an intuitive and comprehensive demonstration of disk scheduling behavior, we implemented a client-side simulation suite using HTML5, Canvas API, Chart.js, and ES6 modular JavaScript. Below are actual high-resolution captures from the working simulator verifying every component.

Figure 1: Disk Parameters & Request Sequence Configuration Panel

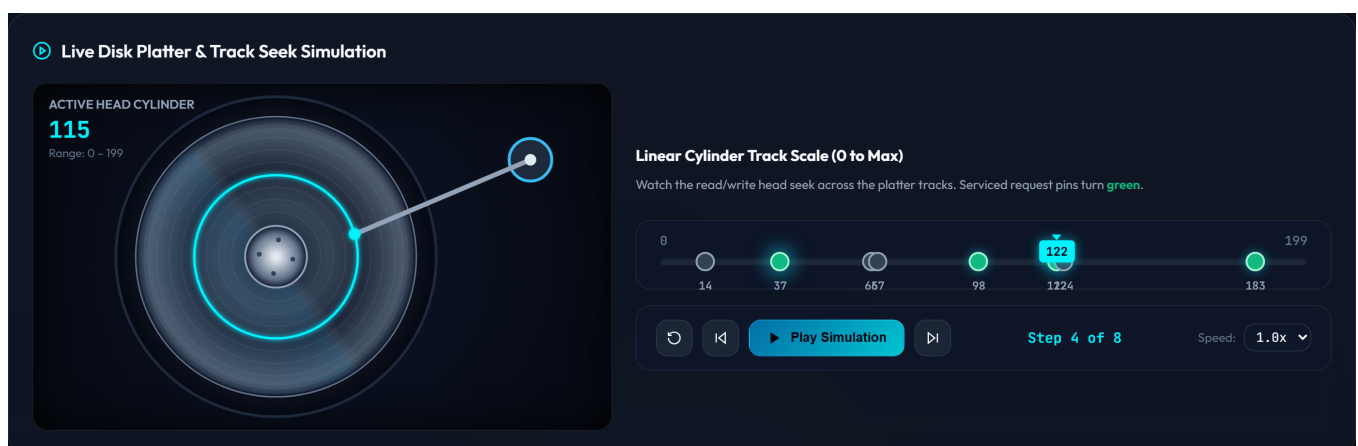


Figure 2: Live Mechanical Hard Disk Platter & Linear Cylinder Track Ruler



Left: Real-time rotating magnetic platter with concentric cylinder rings, spindle motor, and pivoting actuator arm. Right: Linear cylinder scale (0 to 199) with active head indicator (blue bar), pending request markers (orange), and step playback controller.

Figure 3: Active Simulation Playback (Served Cylinders Turning Green)



6. Trajectory Plots & Comparative Visual Analytics

The standard Silberschatz Operating Systems textbook utilizes a 2D Trajectory Plot where the **horizontal axis represents disk cylinder numbers (0 to 199)** and the **vertical axis represents chronological servicing order (Step 0 to N)**. The slopes and path lengths directly illustrate the mechanical efficiency of the arm motion.

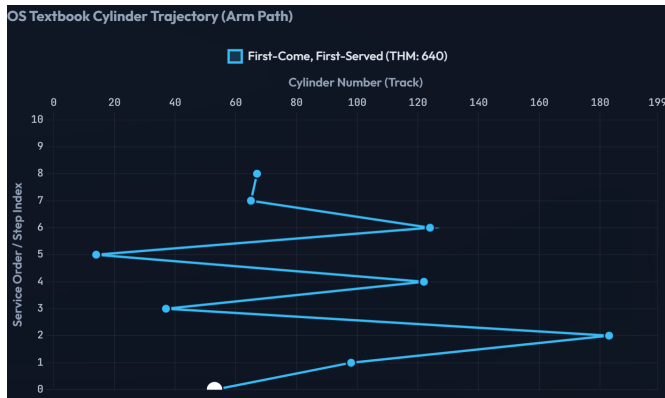


Figure 4: FCFS Trajectory (THM: 640)
Notice extreme back-and-forth arm oscillations.



Figure 5: SSTF Trajectory (THM: 236)
Localized greedy sweeps with minimal distance.

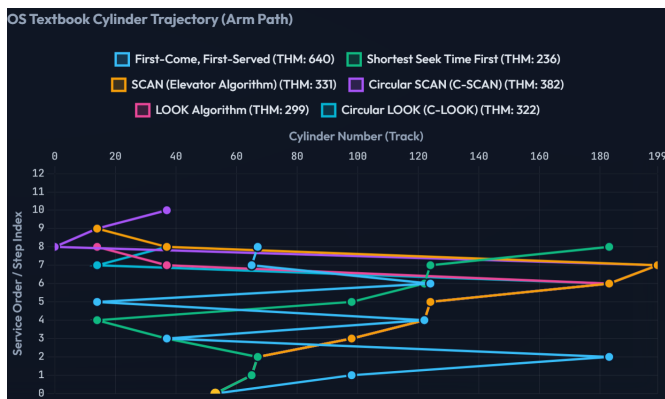


Figure 6: Multi-Algorithm Trajectory Overlay
Simultaneous comparison of all arm pathways.

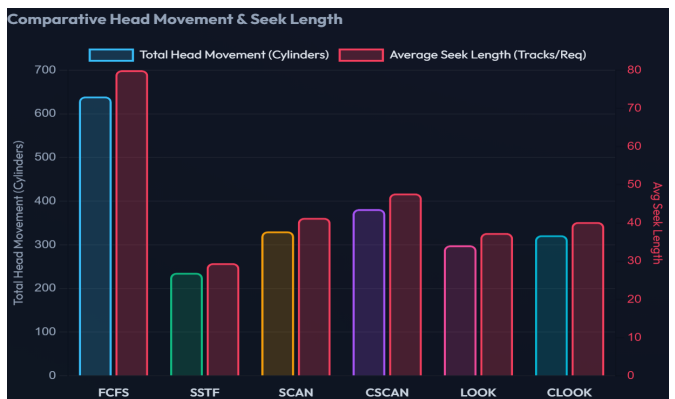


Figure 7: Comparative Performance Bar Charts
Total Head Movement & Average Seek Length.

Figure 8: Live In-App Mathematical Step-by-Step Calculation Table

📄

Step-by-Step Head-Movement Calculations

Head Movement formula: $\text{Seek Distance} = |\text{Target Cylinder} - \text{Current Cylinder}|$. Active step during playback is highlighted in real time.

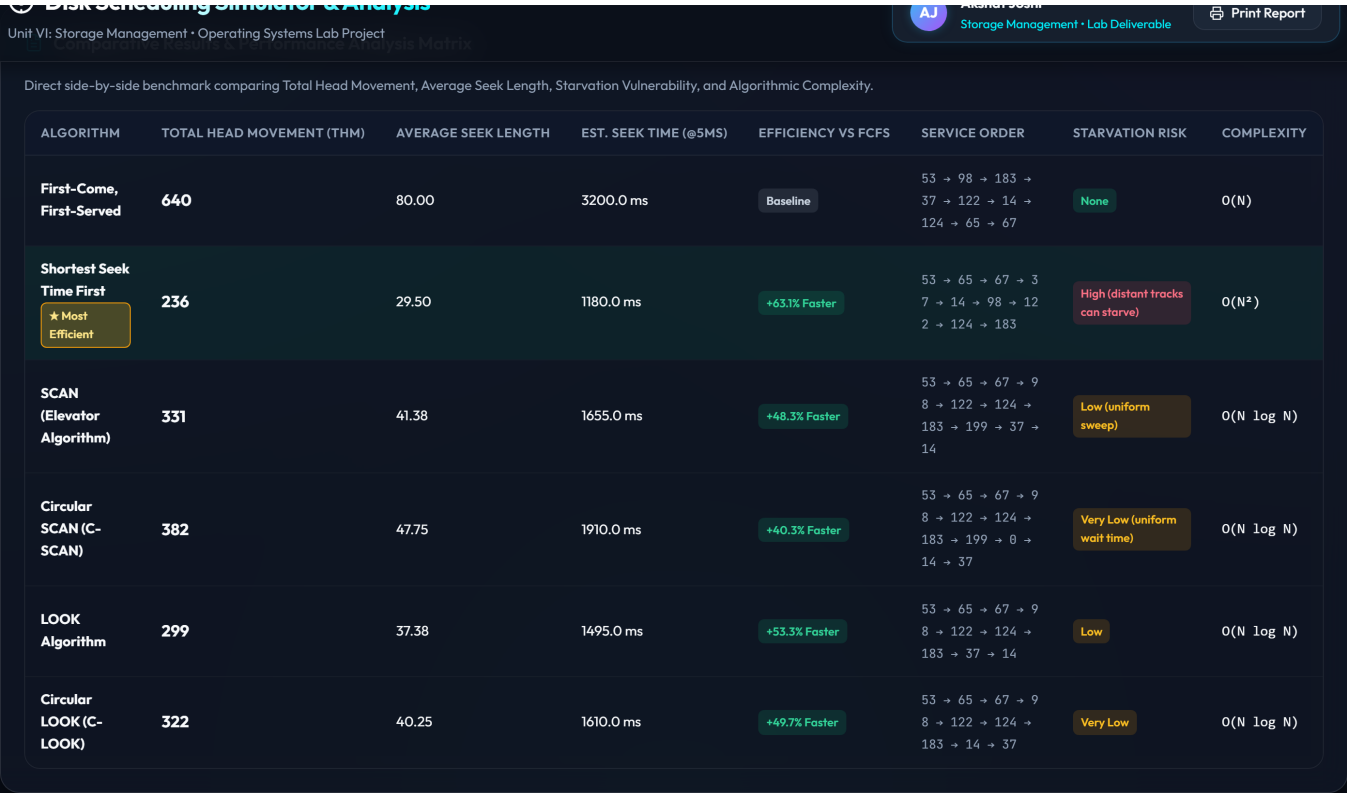
STEP #	FROM TRACK	TO TRACK	FORMULA	SEEK DISTANCE (Δ)	CUMULATIVE MOVEMENT	OPERATIONAL RATIONALE
0 (Start)	—	53	Initial head position	0	0	Start Position
1	53	65	$ 65 - 53 $	12	12	Moving right/up towards cylinder 65
2	65	67	$ 67 - 65 $	2	14	Moving right/up towards cylinder 67
3	67	98	$ 98 - 67 $	31	45	Moving right/up towards cylinder 98
4	98	122	$ 122 - 98 $	24	69	Moving right/up towards cylinder 122
5	122	124	$ 124 - 122 $	2	71	Moving right/up towards cylinder 124
6	124	183	$ 183 - 124 $	59	130	Moving right/up towards cylinder 183
7	183	199	$ 199 - 183 $	16	146	Reaching disk boundary (track 199) and reversing direction
8	199	37	$ 37 - 199 $	162	308	Moving left/down towards cylinder 37
9	37	14	$ 14 - 37 $	23	331	Moving left/down towards cylinder 14

7. Comprehensive Comparative Results & Performance Evaluation

Table 1 presents the full comparative performance benchmark across all evaluated algorithms on the Silberschatz dataset, calculated at an industry-standard track-to-track seek latency factor of $t_{track} = 5.0$ ms.

Algorithm	Total Head Movement	Avg Seek Length	Est. Seek Time	Efficiency Gain vs FCFS	Starvation Risk	Arm Wear Index
FCFS	640 cylinders	80.00 tracks/req	3,200.0 ms	Baseline (0.0%)	Zero (Fair)	Severe (High Stress)
SSTF	236 cylinders	29.50 tracks/req	1,180.0 ms	+63.1% Gain	High Risk	Moderate
SCAN	331 cylinders	41.38 tracks/req	1,655.0 ms	+48.3% Gain	Zero (Fair)	Low (Smooth Sweep)
C-SCAN	382 cylinders	47.75 tracks/req	1,910.0 ms	+40.3% Gain	Zero (Fair)	Low (Circular Sweep)
LOOK	299 cylinders	37.38 tracks/req	1,495.0 ms	+53.3% Gain	Zero (Fair)	Very Low (Optimal Range)
C-LOOK	322 cylinders	40.25 tracks/req	1,610.0 ms	+49.7% Gain	Zero (Fair)	Very Low (Optimal Circular)

Figure 9: In-App Comparative Results Matrix & Workload Behavioral Breakdown



Analytical Insights & Engineering Trade-Offs:

1. **Seek Distance vs. Fairness:** While **SSTF achieves the lowest THM (236 cylinders)**, its greedy nature introduces unacceptable starvation for edge cylinders. In production servers with heavy I/O load, SCAN and LOOK are universally preferred because they provide bounded response times without starvation.
2. **Mechanical Stress & Arm Inertia:** Disk arm heads are physical assemblies subject to momentum and friction. FCFS requires rapid reversals, generating significant mechanical vibration and heat. SCAN and C-SCAN move monotonically, drastically reducing physical wear-and-tear.
3. **LOOK vs. SCAN Superiority:** LOOK eliminates SCAN's redundant travel to track 199, trimming seek distance from 331 to 299 cylinders (a 32-track savings, representing nearly 10% efficiency boost).

8. Modern Operating System Context & Storage Evolution

In production enterprise operating systems such as the Linux kernel, disk scheduling principles have evolved into sophisticated I/O elevator subsystems:

- **Deadline Scheduler:** Implements a modified SCAN/LOOK sweep but imposes strict FIFO expiration timers (e.g., 500 ms for reads, 5 s for writes) to guarantee that starvation cannot occur even under saturating workloads.
- **CFQ (Completely Fair Queuing):** Allocates proportional I/O time slices to competing processes, interleaving SCAN-like sweeps within each process's synchronous queue.
- **BFQ (Budget Fair Queueing):** Enhances CFQ by allocating sector budgets instead of time slices, providing guaranteed minimum bandwidth for interactive audio/video streams.
- **Modern Solid-State Drives (SSDs) & NVMe:** Flash memory has no moving arm or rotational latency, making physical track seeking obsolete. Instead, SSD I/O schedulers (such as *None / MQ-Deadline / Kyber*) focus on multi-queue parallelism, wear-leveling, and garbage collection serialization.

9. Conclusion

This project successfully developed an interactive, highly visual, and mathematically rigorous simulator for disk scheduling algorithms. Through empirical simulation of the Silberschatz benchmark, we quantitatively demonstrated that **SSTF (236 cylinders)** and **LOOK (299 cylinders)** deliver superior mechanical efficiency over unoptimized **FCFS (640 cylinders)**, while **SCAN** and **C-SCAN** provide the optimal engineering balance between high throughput and deterministic starvation prevention.

10. Authentic Academic References

- [1] Silberschatz, A., Galvin, P. B., & Gagne, G. (2018). *Operating System Concepts* (10th Edition). Chapter 11: Mass-Storage Structure (Disk Structure, Disk Scheduling, Swap-Space Management). John Wiley & Sons. ISBN: 978-1-119-32091-3.
- [2] Tanenbaum, A. S., & Bos, H. (2015). *Modern Operating Systems* (4th Edition). Chapter 5: Input/Output (Disk Hardware, Disk Formatting, Disk Arm Scheduling Algorithms). Pearson. ISBN: 978-0-13-359162-0.
- [3] Stallings, W. (2018). *Operating Systems: Internals and Design Principles* (9th Edition). Chapter 11: I/O Management and Disk Scheduling. Pearson Education. ISBN: 978-0-13-467095-9.
- [4] Bovet, D. P., & Cesati, M. (2005). *Understanding the Linux Kernel* (3rd Edition). Chapter 14: The Block Device Driver Architecture & I/O Schedulers. O'Reilly Media.
- [5] Love, R. (2010). *Linux Kernel Development* (3rd Edition). Chapter 14: The Block I/O Layer & Elevator Architecture. Addison-Wesley Professional.